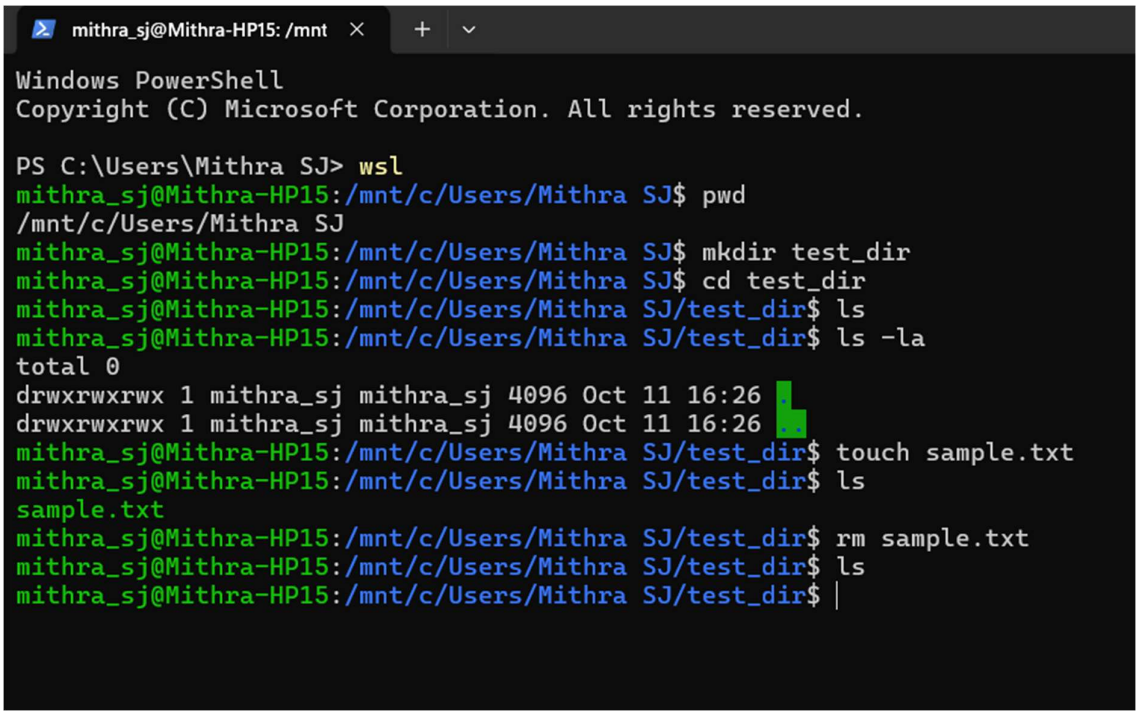
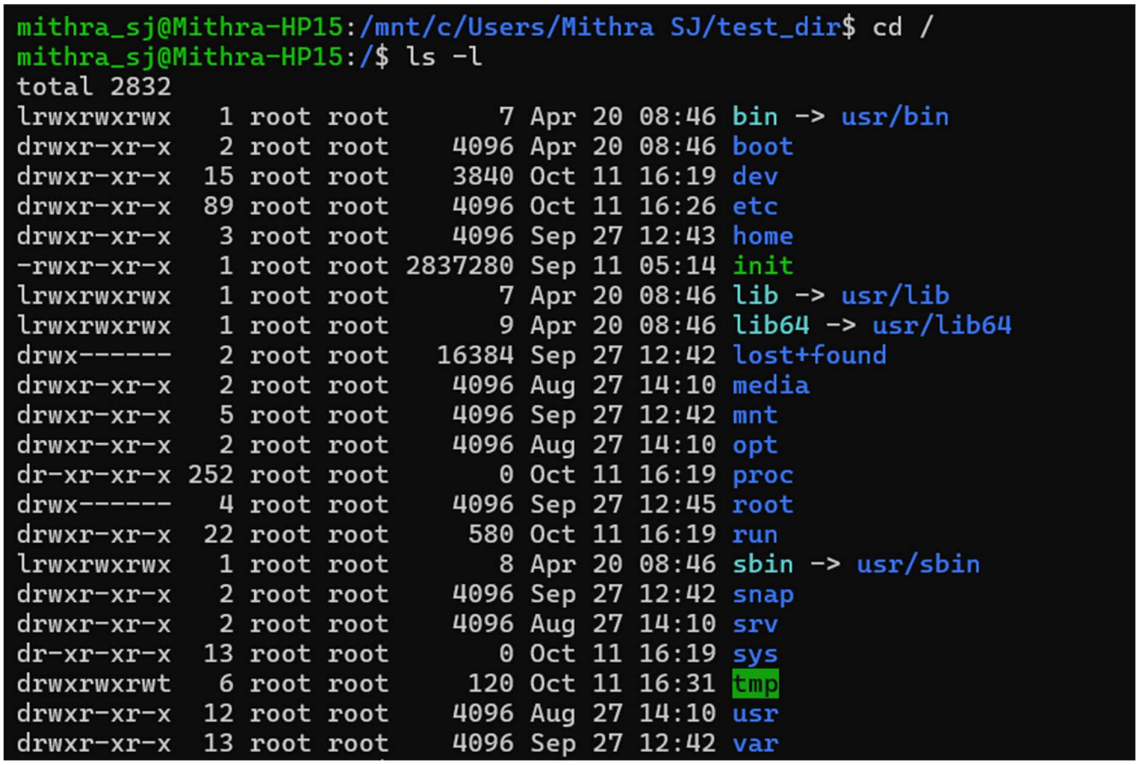
Linux Commands (09-10-2026)

Linux Commands Investigation:



Linux File System Explorer:



```
mithra_sj@Mithra-HP15:/$ ls /etc
PackageKit                    credstore.encrypted  ethertypes  issue.net       mime.types          pm              screenrc          supercat
X11                           cron.d               fonts       kernel          mke2fs.conf         polkit-1        security          sysctl.d
adduser.conf                  cron.daily           fstab       landscape       modprobe.d          ppp             selinux           systemd
alternatives                  cron.hourly          fuse.conf   ld.so.cache     modules             profile         sensors.d         terminfo
apparmor                      cron.monthly         gai.conf    ld.so.conf      modules-load.d      profile.d       sensors3.conf     tmpfiles.d
apparmor.d                    cron.weekly          glvnd       ld.so.conf.d    mtab                protocols       services          ubuntu-advantage
apport                        cron.yearly          gnutls      ldap            nanorc              python3         sgml              ucf.conf
apt                           crontab              gprofng.rc  legal           netconfig           python3.14      shadow            udev
bash.bashrc                   dbus-1               groff       libaudit.conf   netplan             rc0.d           shadow-           update-manager
bash_completion               dconf                group       locale.conf     network             rc1.d           shells            update-motd.d
bash_completion.d             debconf.conf         group-      locale.gen      networkd-dispatcher rc2.d           skel              vconsole.conf
bindresvport.blacklist        debian_version       gshadow     localtime       networks            rc3.d           ssh               vim
binfmt.d                      default              gshadow-    logcheck        newt                rc4.d           ssl               vtrgb
byobu                         deluser.conf         gss         login.defs      nsswitch.conf       rc5.d           subgid            vulkan
ca-certificates               depmod.d             gtk-3.0     logrotate.conf  opt                 rc6.d           subgid-           wgetrc
ca-certificates.conf          dhcp                 host.conf   logrotate.d     os-release          rc5.d           subuid            wsl-distribution.conf
ca-certificates.conf.dpkg-old dhcpcd.conf          hostname    lsb-release     pam.conf            resolv.conf     subuid-           wsl.conf
chrony                        dpkg                 hosts       machine-id      pam.d               rmt             sudo.conf         xattr.conf
cloud                         e2scrub.conf         init.d      magic           passwd              rpc             sudo_logsrvd.conf xdg
console-setup                 environment          inputrc     magic.mime      passwd-             rsyslog.conf    sudoers           xml
credstore                     environment.d        issue       manpath.config  perl                rsyslog.d       sudoers.d         zsh_command_not_found
mithra_sj@Mithra-HP15:/$ ls /home
mithra_sj
mithra_sj@Mithra-HP15:/$
```

Linux Processes:

```
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ps
    PID TTY          TIME CMD
    621 pts/0    00:00:00 bash
    938 pts/0    00:00:00 ps
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24116 15392 ?        Ss   16:19   0:00 /sbin/init
root           2  0.0  0.0   3180  2204 hvc0     Sl+  16:19   0:00 /init
root           6  0.0  0.0   3180  2044 hvc0     Sl+  16:19   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          46  0.0  0.1  50400 16696 ?        S<s  16:19   0:00 /usr/lib/systemd/systemd-journald
systemd+      78  0.0  0.1  22416 14460 ?        Ss   16:19   0:00 /usr/lib/systemd/systemd-resolved
root          85  0.0  0.1  35412 12368 ?        Ss   16:19   0:00 /usr/lib/systemd/systemd-udevd
root         148  0.0  0.0   2888  1948 ?        Ss   16:19   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         149  0.0  0.0   4472  3140 ?        Ss   16:19   0:00 /usr/sbin/cron -f -P
message+     150  0.0  0.0   8800  5444 ?        Ss   16:19   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
root         167  0.0  0.2  44096 29628 ?        Ss   16:19   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         176  0.0  0.0  18436  9436 ?        Ss   16:19   0:00 /usr/lib/systemd/systemd-logind
syslog       204  0.0  0.0 220548  5816 ?        Ssl  16:19   0:00 /usr/sbin/rsyslogd -n -iNONE
root         212  0.0  0.0   5492  2732 tty1     Ss+  16:19   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/u
root         222  0.0  0.2 123460 32820 ?        Ssl  16:19   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
_chrony      226  0.0  0.0  20424  6380 ?        S    16:19   0:00 /usr/sbin/chronyd -n -F 1 -x
_chrony      247  0.0  0.0  12096  2452 ?        S    16:19   0:00 /usr/sbin/chronyd -n -F 1 -x
root         329  0.0  0.0   8648  5556 ?        Ss   16:19   0:00 login -- mithra_sj
mithra_+     372  0.0  0.1  22336 13140 ?        Ss   16:19   0:00 /usr/lib/systemd/systemd --user
mithra_+     374  0.0  0.0  22916  4088 ?        S    16:19   0:00 (sd-pam)
mithra_+     404  0.0  0.0   6136  5484 pts/1    Ss+  16:19   0:00 -bash
root         619  0.0  0.0   3184  1116 ?        Ss   16:26   0:00 /init
root         620  0.0  0.0   3200  1256 ?        S    16:26   0:00 /init
mithra_+     621  0.0  0.0   6268  5624 pts/0    Ss   16:26   0:00 -bash
mithra_+     948  0.0  0.0   7160  4388 pts/0    R+   16:44   0:00 ps aux
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ top
top - 16:44:32 up 25 min,  1 user,  load average: 0.01, 0.03, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  11782.6 total,  11200.5 free,    568.0 used,    183.0 buff/cache
MiB Swap:   3072.0 total,   3072.0 free,      0.0 used.  11214.6 avail Mem
```

```
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ top
top - 16:44:32 up 25 min,  1 user,  load average: 0.01, 0.03, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  11782.6 total,  11200.5 free,    568.0 used,    183.0 buff/cache
MiB Swap:   3072.0 total,   3072.0 free,      0.0 used.  11214.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24116  15392  11612 S   0.0   0.1   0:00.49 systemd
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     46 root      19  -1   50400  16696  15416 S   0.0   0.1   0:00.16 systemd-journal
     78 systemd+  20   0   22416  14460  11968 S   0.0   0.1   0:00.07 systemd-resolve
     85 root      20   0   35412  12368   9232 S   0.0   0.1   0:00.48 systemd-udevd
    148 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.04 chronyd-starter
    149 root      20   0    4472   3140   2892 S   0.0   0.0   0:00.00 cron
    150 message+  20   0    8800   5444   4760 S   0.0   0.0   0:00.04 dbus-daemon
    167 root      20   0   44096  29628  14492 S   0.0   0.2   0:00.19 networkd-dispat
    176 root      20   0   18436   9436   8192 S   0.0   0.1   0:00.07 systemd-logind
    204 syslog    20   0  220548   5816   4516 S   0.0   0.0   0:00.05 rsyslogd
    212 root      20   0    5492   2732   2548 S   0.0   0.0   0:00.01 agetty
    222 root      20   0  123460  32820  18212 S   0.0   0.3   0:00.14 unattended-upgr
    226 _chrony   20   0   20424   6380   5608 S   0.0   0.1   0:00.07 chronyd
    247 _chrony   20   0   12096   2452   1812 S   0.0   0.0   0:00.00 chronyd
    329 root      20   0    8648   5556   4712 S   0.0   0.0   0:00.01 login
    372 mithra_+  20   0   22336  13140  10864 S   0.0   0.1   0:00.07 systemd
    374 mithra_+  20   0   22916   4088   2088 S   0.0   0.0   0:00.00 (sd-pam)
    404 mithra_+  20   0    6136   5484   3928 S   0.0   0.0   0:00.01 bash
    619 root      20   0    3184   1116    980 S   0.0   0.0   0:00.00 SessionLeader
    620 root      20   0    3200   1256   1108 S   0.0   0.0   0:00.06 Relay(621)
    621 mithra_+  20   0    6268   5624   3928 S   0.0   0.0   0:00.14 bash
    958 mithra_+  20   0    8172   6052   3948 R   0.0   0.1   0:00.01 top
```

Linux Permissions:

```
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ touch perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l perm_text.txt
ls: cannot access 'perm_text.txt': No such file or directory
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ chmod perm_test.txt
chmod: missing operand
Try 'chmod --help' for more information.
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l perm_test.txt
-rwxrwxrwx 1 mithra_sj mithra_sj 0 Oct 11 16:36 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ chmod 755 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l
total 0
-rwxrwxrwx 1 mithra_sj mithra_sj 0 Oct 11 16:36 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ chmod 644 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l
total 0
-rwxrwxrwx 1 mithra_sj mithra_sj 0 Oct 11 16:36 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ chmod 777 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l
total 0
-rwxrwxrwx 1 mithra_sj mithra_sj 0 Oct 11 16:36 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$ ls -l
total 0
-rwxrwxrwx 1 mithra_sj mithra_sj 0 Oct 11 16:36 perm_test.txt
mithra_sj@Mithra-HP15:/mnt/c/Users/Mithra SJ/test_dir$
```